Prompt Engineering

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Outline of the Lecture

- ☞ Refresher: GenAI
- ☞ What are prompts
- ☞ What is prompt engineering
- ☞ Anatomy of prompt
- ☞ Effective prompting
- ☞ Context Engineering
- ☞ Summary

Refresher: What is Generative AI

- ☞ (Emphasis on aspects relevant to prompt Engineering)
- ☞ A newer, better version of AI (based on LLMs)
- ☞ Learns and evolves faster and further
- ☞ Unprecedented adapting capabilities
- ☞ Generates novel content, using training data

Generative AI: Uses

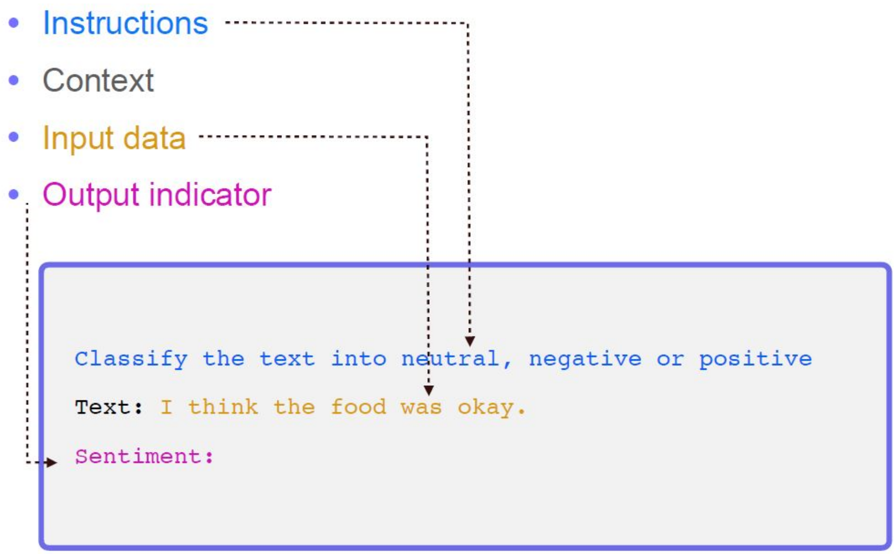
- ☞ Explaining and teaching
- ☞ Writing and rewriting
- ☞ Analyzing and summarizing:
  (especially valuable for statisticians)
- ☞ Organizing
- ☞ Creative brainstorming

Interacting with AI

- ☞ You know how LLMs work
- ☞ We do not need to create new one
- ☞ But we need to know how to existing ones
- ☞ We need to interact with AI
- ☞ The primary way that we interact with AI is through a text entry box

Interacting with AI

☞ What we type in that box is called **prompting**.



What are Prompts

Chat GPT

Fast answer

A **prompt** is the instruction or input you give to an AI system (like ChatGPT) to tell it what you want it to do.

Think of a prompt as:

- a question,
- a command,
- or a description of a task.

Gemini

AI Overview

prompts

A **prompt** is a natural language request, instruction, or query given to an artificial intelligence tool (such as ChatGPT, Copilot, or Gemini) to guide its response. Think of it like giving directions to a highly capable personal assistant; clear, detailed prompts yield the best results. University of Lethbridge

What are Prompts

- ☞ A prompt is a natural language request submitted to a language model to receive a response
- ☞ **Prompts** involve instructions and context passed to a language model to achieve a desired task
- ☞ Prompts can contain questions, instructions, contextual information, short examples, and partial input for the model to complete or continue.

What are Prompts

- Think of a prompt as:
- a question
- a command
- or a description of a task
- Even an example
- A prompt is a natural language request submitted to a language model to receive a response

Why do we need it?

- ☞ LLMs and GenAI do not “think” like humans, even though they attempt to mimic humans.
- ☞ They requires detailed instructions to create relevant high-quality outputs
- ☞ Prompts work by giving the AI a sequence of text that it uses to predict what text should come next.
- ☞ AI generate responses by finding patterns in language learned from enormous amounts of text.

Types of Prompts: Examples

- **Instruction prompts** — "Translate this to Spanish."
- **Creative prompts** — "Invent a fantasy world."
- **Question prompts** — "Why is the sky blue?"
- **Role prompts** — "Act as a fitness coach.
- **Few-shot prompts** — giving examples before asking for a result.

Prompts: Good Versus Bad

Example:

“Explain regression.”

- ☞ Good or bad?
- ☞ Problems?
- ☞ Too broad

“Explain linear regression to MPH students with a simple public health example and minimal math.”

- ☞ Audience defined
- ☞ Complexity adjusted
- ☞ Context provided

Common Problems in Bad Prompts

- ☞ Too vague
- ☞ Audience missing or ill-defined
- ☞ No format
- ☞ No Constraints
- ☞ Ambiguous goals

So, we need learn how to “talk” to AI effectively. The discipline that teaches you that is Prompt Engineering.

What is Prompt Engineering?

Chat GPT

what is prompt engineering?

Prompt engineering is the process of structuring, designing, and refining text instructions (prompts) to ensure generative AI models produce accurate, reliable, and high-quality outputs. Because large language models (LLMs) operate by predicting the next word rather than understanding intent, precise input phrasing directly determines the quality of the result. Oracle +3

It bridges the gap between human language and machine execution, transforming general AI systems into highly specialized tools. IBM +1

Gemini

what is prompt engineering

Fast answer

Prompt engineering is the practice of designing and refining the instructions you give to an AI model so it produces better, more accurate, or more useful outputs.

In simple terms: it's learning how to "talk" to AI effectively.

What is Prompt Engineering

- ☞ By carefully crafting the wording and context, users can guide AI systems to produce highly **accurate** and **relevant** outputs

- ☞ Prompt engineering is the practice of developing,, testing, refining and optimizing the text or instructions given to generative AI models.

Why learn prompt engineering?

- Important for research, discoveries, and advancement
- Helps to test and evaluate the limitations of LLMs and Gen AI
- Enables all kinds of innovative applications
- Lucrative career prospects ????

Why learn prompt engineering?

Is Prompt Engineering a Career?

Yes. Skills in prompt engineering are useful for:

☞ AI product design

☞ automation

☞ chatbot development

☞ AI-assisted programming

☞ workflow optimization

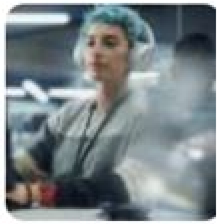
Though many experts now see it as part of broader “AI interaction design” rather than a standalone long-term profession

Prompt Engineering: Career

Forbes

The Hot, New High-Paying Career Is An AI Prompt Engineer

As AI job listings have surged 42%, prompt engineering is quickly becoming a hot job with the ascendancy of artificial intelligence.



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The Hottest AI Job of 2023 Is Already Obsolete

Prompt engineering, a role aimed at crafting the perfect input to send to a large language model, was poised to become one of the hottest jobs in artificial intelligence. What happened?

Anatomy of a Prompt

Though prompts can get very complex, they tend to boil down to a few components:

- ☞ **Task**: What you want the AI to do
- ☞ **Instructions**: How you want the AI to do it
- ☞ **Context**: What you want the AI to know
- ☞ **Format:** How you want the answer structured
- ☞ **Constraints:** What rules, limit and boundaries you want to set
- ☞ **Creativity**: How creative you want AP to be?

TIC or 3 Cs?

☞ **Task, Instructions, Context:**

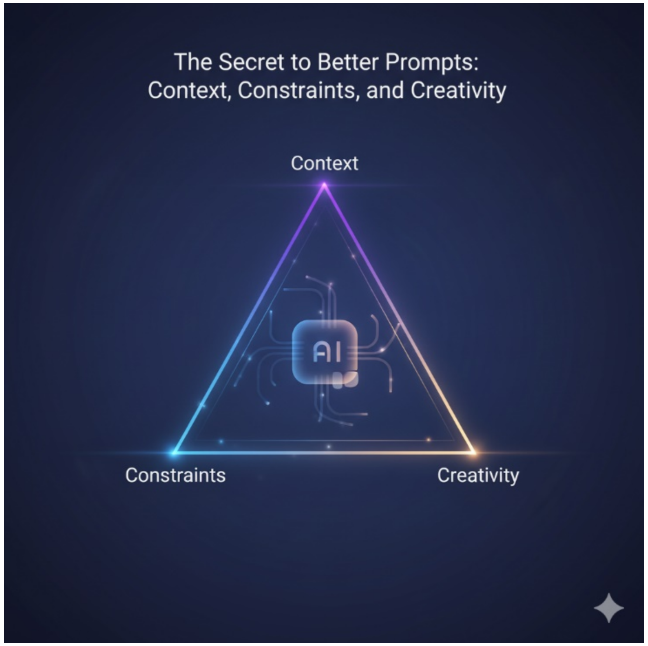
Profs. Goel, Levy, and Svoronos
Harvard Kennedy School

(common acronym TIC*) *(https://generative-ai-course.hks.harvard.edu/)

☞ **Context**

☞ **Constraints**

☞ **Creativity**



[ ]: Task

( ): Instructions

{ }: Context

Writing an admissions essay

I am applying to the online masters of public health program in Keck school of medicine. I would like you to produce a first draft of one of my essays (300word limit). The essay prompt is the following: "Describe a time when interactions with others and/or an experience caused you to change your mind or expanded your point of view." I would like you to write about my experience changing my mind about the potential of public health. I used to believe that it did not have much potential and now I believe that it has. An interaction with two faculty members and 3 students made me realize that I was wrong and they were right. Please produce an great essay with personal touch but don’t ask me any questions; just write the essay.

[ ]: Task

( ): Instructions

{ }: Context

Sometimes helpful: rearranging

Task: I will give you an essay prompt and you will write a first draft of an admissions essay.

Instructions: Produce an essay responding to the essay prompt, "Describe a time when interactions with others and/or an experience caused you to change your mind or expanded your point of view." It should be personal, crisp and persuasive. Please don’t ask me any questions; just write the essay. Keep it under 300 words.

Context: I am applying to the Keck School of Medicine’s online masters of public health program. I used to believe that online learning did not have the potential to transform online education. A conversation with two faculty members and three students made me realize I was wrong and they were right. [Add additional context]

Effective prompting: General guidelines

1. Start simple
2. Gradually build (via conversing)
3. Be clear and specific about what you want
4. Provide context
5. Define the desired output format
6. Assign a role (Personas)
7. Specify constraints
8. Use examples
9. Iterative Improvement

Effective prompting: Advanced

- ☞ Chain of thought prompting
- ☞ Decomposition prompting
- ☞ Self-critique/reflection prompting
- ☞ Comparative Prompting
- ☞ Reproducibility Testing
- ☞ Prompt Chaining
- ☞ Retrieval-Augmented Prompting
- ☞ Multi-Perspective Prompting

& others….

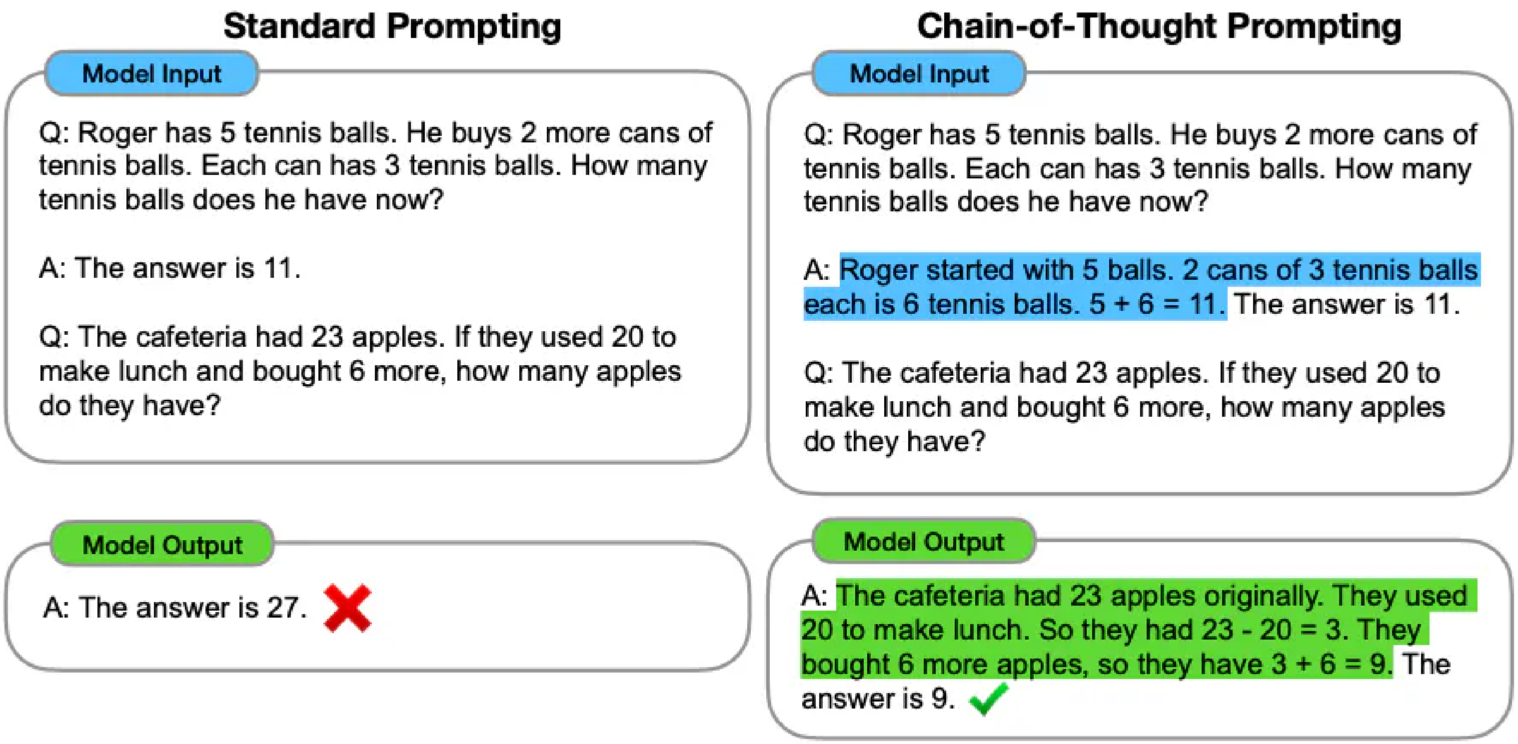
Chain of thought prompting

- Encourages intermediate reasoning before final answers
- Prompts that instruct an LLM to break down their responses into intermediate steps before arriving at an answer.
- **Example:** “Before answering, explain the reasoning process used to determine whether confounding is present.”

Useful for:

- statistics
- study design
- causal reasoning

Chain of thought (CoT)



Source: PAL: <u>Program-aided Language Models</u>

Decomposition Prompting

☞ Break a large task into smaller tasks.

☞ **Weak Prompt:** "Write a grant proposal."

☞ **Better Prompt:** "First generate study aims. Then generate hypotheses. Then propose statistical analyses."

☞ This improves:

-organization

-accuracy

-controllability

Self-Critique / Reflection Prompting

- ☞ Ask the AI to evaluate its own answer.
- ☞ Example: after analysis ask:
  "Identify possible statistical weaknesses in this analysis."
- ☞ Very useful for:

-academic writing

-code review

-study design

Comparative Prompting:

- ☞ Generate multiple alternatives and compare them.
- ☞ This reveals: tone differences, complexity adaptation,

communication quality

- ☞ Example: “Generate three different ways to explain p-values:

1) for undergraduate students

2) for MPH students

3)for biostatistics PhD students.”

Reproducibility Testing

- ☞ Same prompt → different outputs
- ☞ **In-class Exercise Example**
- ☞ Use the exact same prompt three times in three platforms: "Write a short public health message encouraging flu vaccination."
- ☞ Then compare:

Wording, framing, tone, emphasis, factual consistency

- ☞ Key lesson:
  AI outputs are probabilistic, not deterministic.

Prompt Chaining

☞ Outputs from one prompt become inputs for another.

☞ **Example Workflow**

-Generate research question

-Generate hypotheses

-Generate statistical plan

-Generate limitations

-Generate presentation slides

This mimics real analytical pipelines.

Retrieval-Augmented Prompting

- ☞ Provide external information directly in the prompt.
- ☞ **Example:** “Using the following CDC summary, create a 200-word policy brief.”
- ☞ This improves:

factual grounding

accuracy

consistency with source material

Multi-Perspective Prompting

- ☞ Ask the AI to analyze from several viewpoints.
- ☞ Example: “Discuss mandatory vaccination policies from:

1) an epidemiologist’s perspective

2) a policymaker’s perspective

3) a patient advocate’s perspective”

- ☞ Excellent for: classroom discussion, ethics, policy analysis

Remember

- ☞ Effective prompting is not about memorizing tricks.

It is about:

- ☞ clear communication
- ☞ structured thinking
- ☞ iterative refinement
- ☞ understanding task design

In many ways good prompting is similar to a good scientific writing or good statistical report. So, naturally it fits in epidemiology, biostatistics and public health.

A Simple Prompt Formula

☞ A very effective beginner template:

☞ **Role + Task + Context + Format + Constraints**

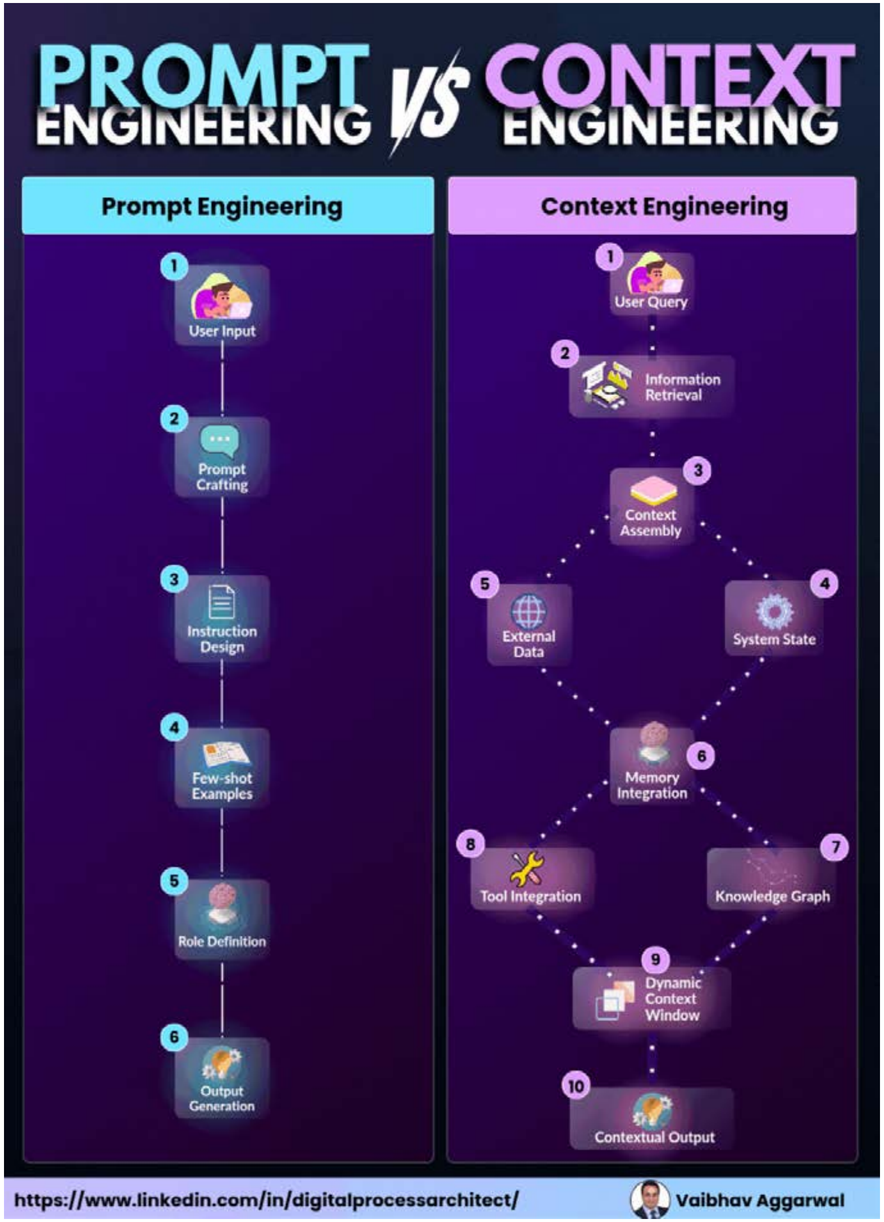
☞ Context is very important

☞ New emerging field: Context Engineering

Prompt Engineering VS Context Engineering

- ☞ Prompt engineering is how you talk to the AI (the instructions, tone, and formatting).
- ☞ Context engineering is what the AI knows before it even starts responding (the data, tools, and background environment).
- ☞ While prompt engineering sets the rules, context engineering builds the ecosystem that makes those rules

Prompt Engineering VS Context Engineering



Prompt Engineering VS Context Engineering

- Prompt engineering optimizes one interaction.
- Context engineering thinks in sequences:
  What did previous turns establish?
  What tool outputs carry forward?
  What should persist three steps from now?

References

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- ☞ https://github.com/dair-ai/Prompt-Engineering-Guide
- ☞ *PAL: Program-aided Language Models*
- ☞ https://generative-ai-course.hks.harvard.edu/
- ☞ ChatGPT (OpenAI)
- ☞ Gemini